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TRANSFORM YOURSELF

Department of Computer Science and Engineering

24GEP31 - MINI PROJECT

REVIEW–2

Smart Public Transportation Route Optimization

SDG 9 – Industry, Innovation and Infrastructure

SDG 11 – Sustainable Cities and Communities

SDG 13 – Climate Action.

TEAM MEMBERS DETAILS:
25CSR253 – SAKTHIVEL S
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Mentor Details:
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Assistant Professor / CSE

Problem Definition & Scope

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Problem Statement:

Static municipal bus timetables fail during stochastic peak-hour passenger surges, causing severe curb overcrowding, increased commuter wait times, and high carbon emissions. Manual dispatching lacks real-time corridor passenger counts and optimal routing.

Key Project Objectives:

- Prioritize unserved commuter clusters using volume-based binary max-heap ranking.
- Consolidate same-corridor passenger requests via FIFO queues to prevent redundant dispatches.
- Compute optimal two-segment routes (Depot → Pickup → Destination) via Dijkstra's algorithm.
- Allocate fleet based on best-fit capacity (MiniBus vs. StandardBus) with live seat deduction.
- Maintain real-time bidirectional synchronization with MySQL database via phpMyAdmin.

Project Scope:

- Vetrinagar network: 13 halts (1 Central Depot D00 + 12 passenger stations) & 44 road corridors.
- Fleet of 4 operational buses across 2 categories (30 to 50 seat capacities).
- Demand-based relief-trip planning with live traffic and road closure detours.

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Objectives & Expected Outcomes

Core Technical Objectives:

- Dynamic Surge Prioritization: Max-heap PriorityQueue ranking by volume (DESC) and arrival time (ASC).
- Same-Corridor Demand Aggregation: ArrayDeque FIFO sub-queues consolidating multiple commuter bookings.
- Optimal Pathfinding: Dijkstra algorithm computing minimum travel times with congestion penalties.
- Best-Fit Fleet Selection: Matching surge volume to smallest sufficient vehicle, tracking live seats.
- Bidirectional Persistence: Sub-3.0s JDBC synchronization daemon linking Java backend with MySQL.

Expected Algorithmic Workflow:

PASSENGER DEMAND ↓ CORRIDOR AGGREGATION (ArrayDeque) ↓ HEAP PRIORITIZATION (PriorityQueue) ↓ BEST-FIT BUS ALLOCATION ↓ TWO-SEGMENT DIJKSTRA ROUTING ↓ RELIEF TRIP DISPATCH ↓ DATABASE UPDATE (MySQL)

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Technical Approach & Feasibility

Java OOP Implementation

- Encapsulation: Bus, Stop, Road, DemandBatch protect mutable states.
- Inheritance: Abstract Bus extended by MiniBus (30) & StandardBus (40-50).
- Polymorphism: Specialized fuel calculations & corridor width restrictions.
- Abstraction: Decoupled Router & Persistence interfaces.
- Composition: WeightedGraph encapsulates Road lists; Service links heap & fleet.

Structure	Algorithmic Purpose in Project
PriorityQueue	Binary Max-Heap for real-time demand surge prioritization
HashMap	O(1) stop lookup, corridor mapping, and graph adjacency
ArrayDeque	FIFO sub-queue for same-corridor passenger batch aggregation
WeightedGraph	Adjacency list representation of 13 halts & 44 directed road corridors
ArrayList	Dynamic fleet inventory, dispatched relief missions & audit log

Technology Stack (Java 21 + MySQL)

Component	Actual Project Technology Used
Programming Language	Java 21 LTS (Strong Typing, Records, High Performance)
Web Server Architecture	com.sun.net.httpserver (Built-in Lightweight Java REST API)
Algorithmic Engine	Dijkstra Router, Binary Max-Heap, FIFO Corridor Batching
Frontend Dashboard	Pure HTML5, Modern CSS3, Interactive Vector SVG Canvas, JS
Database Engine	MariaDB / MySQL Server (InnoDB Engine, ACID Transactions)
Database Connectivity	JDBC Connector (mysql-connector-j-9.2.0.jar) + phpMyAdmin

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System Architecture / Methodology

SYSTEM ARCHITECTURE

COMMUTER / SUPERVISOR

↓

WEB DASHBOARD (HTML5 / SVG Vector Canvas)

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HTTP REST API (Java 21 com.sun.net.httpserver)

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CORRIDOR DEMAND AGGREGATION (ArrayDeque)

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DEMAND PRIORITIZATION (Binary Max-Heap PriorityQueue)

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BEST-FIT FLEET ALLOCATION & CAPACITY DEDUCTION

↓

TWO-SEGMENT DIJKSTRA ROUTE PLANNER

↓

RELIEF TRIP DISPATCH & PERSISTENT MYSQL UPDATE

System Flow

Input:

- Commuter requests (Origin, Destination, Passenger Count).
- Fleet inventory, current halt locations, and available capacities.
- Road conditions, live travel times, and closure flags.

Process:

- Same-corridor aggregation (FIFO) & volume-based max-heap ranking.
- Best-fit vehicle matching (MiniBus vs. StandardBus).
- Dijkstra two-stage pathfinding avoiding traffic & closed corridors.
- Dynamic seat capacity deduction upon passenger boarding.

Output:

- Prioritized demand batches & optimal two-segment route turn-by-turn.
- Live SVG map rendering with pulsing crowd badges ('Pax: 25').
- Persistent MySQL trip records & verified carbon savings (SDG 13).

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Project Planning & Timeline

Week 1 — Problem Analysis & Graph Modeling

• Formulated municipal transit bottlenecks and aligned with SDG 9, 11, and 13.
• Formalized the 13-node, 44-edge weighted directed graph of Vetrinagar municipality.
• Designed Java 21 OOP class hierarchy (Bus, MiniBus, StandardBus, Stop, Road).
• Established automated JUnit test scaffolding for graph traversal and edge validation.

Week 2 — Core Algorithmic Engines Development

• Implemented DijkstraRouter using min-heap priority queues for two-segment pathfinding.
• Engineered DemandBatch aggregator using ArrayDeque FIFO sub-queues for same-halt requests.
• Built Binary Max-Heap PriorityQueue ranking surges by volume (DESC) and timestamp (ASC).
• Developed best-fit fleet capacity matcher and dynamic remaining capacity tracker.

Review 2 — Web Interface, Routing Visualization & Database Integration

• Developed interactive 840x620 SVG transit canvas with live crowd badges and colored roads.
• Implemented lightweight Java HTTP REST API server using com.sun.net.httpserver.
• Designed normalized MySQL database schema (transport_optimization) across 5 core tables.
• Engineered DatabaseManager with 3.0s bidirectional synchronization daemon with phpMyAdmin.

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Implementation Status & Review-2 Progress

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Completed & Demonstrated Modules:

- Interactive SVG Transit Dashboard with live bus tracking and dynamic crowd badges.
- Same-Corridor Demand Aggregation Engine (ArrayDeque) & Max-Heap Prioritizer (PriorityQueue).
- Two-Segment Dijkstra Routing Engine with automated detours around road closures.
- Best-Fit Fleet Allocation Engine with dynamic remaining capacity deduction.
- MySQL Database persistence with 3.0s bidirectional sync daemon via phpMyAdmin.
- Passenger Booking Portal & Supervisor Dispatch Control Panel with SDG 13 carbon audit.

Verified System Functionality (Live Review-2 Demos):

- Multi-passenger same-halt consolidation (combining 5 and 20 pax into single 25-pax batch).
- Sub-millisecond PriorityQueue ranking and smallest sufficient bus allocation.
- Dijkstra route computation across 44 corridors executing in 1.42 ms average latency.
- Dynamic seat availability deduction from 30 seats to 5 seats upon passenger boarding.
- Live database edits in phpMyAdmin dynamically updating Java runtime within 3.0 seconds.

Future Enhancements / Next Milestones:

- Integration with GTFS-Realtime live GPS vehicle telemetry feeds.
- Machine learning passenger surge forecasting using Long Short-Term Memory (LSTM) networks.
- Multi-depot collaborative fleet balancing across distributed municipal zones.

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